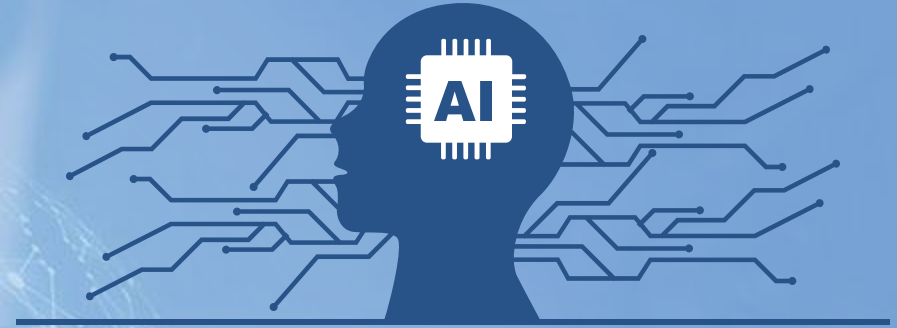




Homework #1

Object Detection



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TOPIC: Object Detection for Group-housed Swine

- Input: 2D RGB image
- Task: object detection





Submission Deadline

2025/10/10 23:59

Kaggle: Hard deadline, No extensions

NTU COOL: Late Submission Policy is Applicable

(See TA01-Assignments_Overview.pdf)

- **We have separately created Kaggle competitions exclusively for NTU students and exclusively for TAICA**
- **Please only participate in the competition designated for you (NTU or TAICA)**
- **The grading policy for TAICA students should be determined by your institution's teaching assistants or professors. Any questions, please contact them.**



Dataset Description

- Download link: On Kaggle competition
- Training set: 1266 images (including img/ & gt.txt)

Each line in gt.txt contains:

```
<frame>, <bb_left>, <bb_top>, <bb_width>, <bb_height>
```

- Testing set: 1864 images (including img/)
- **DO NOT** try to find the ground truth of testing set
- **DO NOT** use testing data during training
- Violating the rules on this page will result in a score of zero.
- If you are uncertain about the legitimacy of the usage, email the TAs for clarification



Evaluation Metric

- Average Precision $mAP_{50:95}$
 - Since we only have one class, it equals to $AP_{50:95}$.
 - The average of the mean average precision calculated at varying IoU thresholds, ranging from 0.50 to 0.95.
 - Please refer to the course slides or this [intro](#).
- There are one simple baseline and one strong baseline, beat them to get the higher score.

➤ Kaggle Competition (testing set) (90%) (according to **private leaderboard**)

μ, σ are calculated on $\text{mAP}_{50:95} \geq \text{private baseline}$.

Your $\text{mAP}_{50:95}$	Points
$\text{mAP}_{50:95} \geq \mu + 2\sigma$	90
$\text{mAP}_{50:95} \geq \mu + \sigma$	80
$\text{mAP}_{50:95} \geq \mu$	75
$\text{mAP}_{50:95} \geq \mu - \sigma$	70
$\text{mAP}_{50:95} \geq \mu - 2\sigma$	65
$\text{mAP}_{50:95} \geq \text{strong baseline}$	60
$\text{mAP}_{50:95} \geq \text{simple baseline}$	Linear between 50 ~ 60

➤ Report (10%)



Kaggle (90%) (For **NTU** Students)

- Use this [link](#) to participate in the competition.
- **DO** rename your team name to <student-id> (e.g., R12345678).
- The maximum daily submissions is 5.
- The public leaderboard shows the score of only 50% test data. Your final score is evaluated on the other 50% test data.
- You can optionally select 2 submissions as the final submissions.
- **DO NOT** use more than one account to participate in the Kaggle competition.
- Violating the rules on this page will result in a score of zero.
- Students of TAICA DO NOT participate in this competition



Kaggle (90%) (For **TAICA** Students)

- Use this [link](#) to participate in the competition.
 - **DO** rename your team name to <**school**_student-id> (e.g., **NYCU_R12345678**).
 - The maximum daily submissions is 5.
 - The public leaderboard shows the score of only 50% test data. Your final score is evaluated on the other 50% test data.
 - You can optionally select 2 submissions as the final submissions.
 - **DO NOT** use more than one account to participate in the Kaggle competition.
-
- Violating the rules on this page will result in a score of zero.
 - Students of NTU DO NOT participate in this competition

- Save predictions in csv format and submit to Kaggle:

```
Image_ID,PredictionString
1,<conf_1> <bb_left_1> <bb_top_1> <bb_width_1> <bb_height_1> <class_1> <conf_2>...
2,<conf_1> <bb_left_1> <bb_top_1> <bb_width_1> <bb_height_1> <class_1> <conf_2>...
...
```

- Image_ID is the index of images in the test set.
- PredictionString contains ALL DENORMALIZED predictions of the corresponding image.
- Use “,” to separate Image_ID and PredictionString **WITHOUT SPACE**.
- Use “**SPACE**” to separate the attributes of bounding boxes.
- Set <class> to **0** to represent pigs.



Programming Spec

- Use Python ≥ 3.10 (for consistency and reproducibility).
 - No loading of pretrained weights (except as feature extractors).
 - Only use the provided dataset — no extra datasets allowed.
 - No plagiarism (Online resources may be consulted. However, using the same code source as classmates may be considered plagiarism).
-
- Violating the above rules on this page will result in a score of zero.
 - If you are uncertain about the legitimacy of the usage, email the TAs for clarification

1. Model Description

- Introduce your model (must include an **architecture illustration & any modifications**)

2. Implementation Details

- Preprocessing, augmentation, hyperparameters, loss functions, training strategies, etc.

3. Result Analysis

- Quantitative improvements (tables, metrics, discussion)
- Visualizations (e.g., example detections, error analysis)

4. Short conclusion

- 3-5 pages (excluding references), exceeding the limit will result in a -5 score



NTU COOL Submission Rules

- Your submission should be a zipped file with the following structure:
 - hw1_<student-id>.zip (e.g., hw1_R12345678.zip)
 - |-- hw1_<student-id> (Should contain this folder, not separate files)
 - |----- report_<student-id>.pdf (Your report)
 - |----- code_< student-id>.zip
 - |----- src/ (Your source code)
 - |----- readme.md (Show how to install the environment, run training & prediction)
 - |----- requirements.txt (The list of necessary packages)
- An incorrect format will result in a deduction of a -5 score.
- Failure to re-implement similar performance will result in a 60% discount of the total score.
- Plagiarism in the report or code will result in 0%.

Any Question

Ask peers first

(Join with name: <school_student-id>)



Then ask TAs

(only for NTU students)

cvpdl.ta.2024@gmail.com

The background of the slide features a person's hand holding a tablet. Overlaid on the tablet is a glowing, wireframe sphere composed of interconnected nodes and lines, resembling a network or data structure. The entire scene is set against a blue gradient background.

Supplementary Material



Download Dataset to Google Drive on Colab

- Kaggle → your profile → Account → Create New API token → upload token to google drive `./kaggle/kaggle.json`.
- Mount google drive to `/content/drive`.

```
from google.colab import drive  
drive.mount('/content/drive')
```

- Use kaggle package to download dataset.

For NTU students

```
! KAGGLE_CONFIGDIR=/content/drive/MyDrive/.kaggle \  
kaggle competitions download \  
-c ntu-cvpdl-2025-hw-1\  
-p ./drive/MyDrive
```

For TAICA students

```
! KAGGLE_CONFIGDIR=/content/drive/MyDrive/.kaggle \  
kaggle competitions download \  
-c taica-cvpdl-2025-hw-1\  
-p ./drive/MyDrive
```




Download Dataset to Google Drive on Colab

- Copy dataset to VM storage and unzip it.

For NTU students

```
! cp ./drive/MyDrive/ntu-cvpdl-2025-hw-1.zip .  
! unzip ./ntu-cvpdl-2025-hw-1.zip -d ./data
```

For TAICA students

```
! cp ./drive/MyDrive/taica-cvpdl-2025-hw-1.zip .  
! unzip ./taica-cvpdl-2025-hw-1.zip -d ./data
```

- Start to read data from path `./data/train` and `./data/test`

- Generate requirements.txt

```
pip freeze > requirements.txt
```

- Install package from requirements.txt

```
pip install -r requirements.txt
```

- Manually create requirements.txt

```
pandas  
numpy==2.2.6  
torch==2.4.0  
torchvision==0.19.0  
...
```



Use Virtual Environment

➤ Create virtual environment for a project and install the needed packages only, for example:

- `virtualenv`
- `conda`
- `pyenv`